Arithmetic of Complex Numbers — Solutions

Warm-up

Question 1a:
Given $z = 2 + 3i$ and $w = 5 + 12i$.

$$\begin{aligned} z + w &= (2 + 3i) + (5 + 12i) \\ &= (2 + 5) + (3 + 12)i \\ &= 7 + 15i \end{aligned}$$

Question 1b:
Given $z = 2 + 3i$ and $w = 5 + 12i$.

$$\begin{aligned} z - w &= (2 + 3i) - (5 + 12i) \\ &= 2 + 3i - 5 - 12i \\ &= (2 - 5) + (3 - 12)i \\ &= -3 - 9i \end{aligned}$$

Question 1c:
Given $z = 2 + 3i$ and $w = 5 + 12i$.

$$\begin{aligned} zw &= (2 + 3i)(5 + 12i) \\ &= 2(5) + 2(12i) + 3i(5) + 3i(12i) \\ &= 10 + 24i + 15i + 36i^2 \\ &= 10 + 39i - 36 \\ &= -26 + 39i \end{aligned}$$

Question 1d:
Given $z = 2 + 3i$ and $w = 5 + 12i$.

To divide complex numbers, we multiply the numerator and denominator by the conjugate of the denominator $\bar{w} = 5 - 12i$.

$$
\begin{aligned}
\frac{z}{w} &= \frac{2+3i}{5+12i} \times \frac{5-12i}{5-12i} \\
&= \frac{(2+3i)(5-12i)}{5^2+12^2} \\
&= \frac{10-24i+15i-36i^2}{25+144} \\
&= \frac{10-9i+36}{169} \\
&= \frac{46-9i}{169} \\
&= \frac{46}{169} - \frac{9}{169}i
\end{aligned}
$$

Question 1e:
Given $z = 2+3i$.

$$
\begin{aligned}
z^2 &= (2+3i)^2 \\
&= 2^2 + 2(2)(3i) + (3i)^2 \\
&= 4 + 12i + 9i^2 \\
&= 4 + 12i - 9 \\
&= -5 + 12i
\end{aligned}
$$

Question 1f:
Given $w = 5+12i$. Let $\sqrt{w} = x + iy$.

$$
\begin{aligned}
(x+iy)^2 &= 5+12i \\
x^2 - y^2 + 2xyi &= 5+12i
\end{aligned}
$$

Equating real and imaginary parts:

$$
\begin{aligned}
x^2 - y^2 = 5 \quad (1) \\
2xy = 12 \implies xy = 6 \quad (2)
\end{aligned}
$$

From (2), $y = \frac{6}{x}$. Substitute into (1):

$$
\begin{aligned}
x^2 - \left(\frac{6}{x}\right)^2 &= 5 \\
x^2 - \frac{36}{x^2} &= 5 \\
(x^2)^2 - 5x^2 - 36 &= 0
\end{aligned}
$$

Let $u = x^2$. Then $u^2 - 5u - 36 = 0 \implies (u-9)(u+4) = 0$. Since x is real, $x^2 = 9$, so $x = \pm 3$.
If $x = 3$, then $y = \frac{6}{3} = 2$. If $x = -3$, then $y = \frac{6}{-3} = -2$.
Thus, $\sqrt{w} = \pm(3+2i)$.

Question 1g:
Given $z = 2 + 3i$.

$$\mathrm{Re}(z) = 2$$

Question 1h:
Given $w = 5 + 12i$.

$$\mathrm{Im}(w) = 12$$

Question 1i:
Given $z = 2 + 3i$.

$$\begin{aligned} |z| &= \sqrt{2^2 + 3^2} \\ &= \sqrt{4 + 9} \\ &= \sqrt{13} \end{aligned}$$

Question 1j:
Given $w = 5 + 12i$.

$$\begin{aligned} |w| &= \sqrt{5^2 + 12^2} \\ &= \sqrt{25 + 144} \\ &= \sqrt{169} \\ &= 13 \end{aligned}$$

Question 2a:
Given $z = 6 - 7i$ and $w = 3 + 4i$.

$$\begin{aligned} \bar{z} &= 6 + 7i \\ \bar{w} &= 3 - 4i \\ \bar{z} + \bar{w} &= (6 + 7i) + (3 - 4i) \\ &= (6 + 3) + (7 - 4)i \\ &= 9 + 3i \end{aligned}$$

Question 2b:
Given $z = 6 - 7i$ and $w = 3 + 4i$.

$$
\begin{aligned}
\bar{w} &= 3 - 4i \\
z - \bar{w} &= (6 - 7i) - (3 - 4i) \\
&= 6 - 7i - 3 + 4i \\
&= (6 - 3) + (-7 + 4)i \\
&= 3 - 3i
\end{aligned}
$$

Question 2c:
Given $z = 6 - 7i$ and $w = 3 + 4i$.

$$
\begin{aligned}
zw &= (6 - 7i)(3 + 4i) \\
&= 6(3) + 6(4i) - 7i(3) - 7i(4i) \\
&= 18 + 24i - 21i - 28i^2 \\
&= 18 + 3i + 28 \\
&= 46 + 3i
\end{aligned}
$$

Question 2d:
Given $z = 6 - 7i$ and $w = 3 + 4i$.

$$
\begin{aligned}
\bar{w} &= 3 - 4i \\
\frac{\bar{w}}{z} &= \frac{3 - 4i}{6 - 7i} \times \frac{6 + 7i}{6 + 7i} \\
&= \frac{(3 - 4i)(6 + 7i)}{6^2 + (-7)^2} \\
&= \frac{18 + 21i - 24i - 28i^2}{36 + 49} \\
&= \frac{18 - 3i + 28}{85} \\
&= \frac{46 - 3i}{85} \\
&= \frac{46}{85} - \frac{3}{85}i
\end{aligned}
$$

Question 2e:
Given $w = 3 + 4i$.

$$
\begin{aligned}
w^2 &= (3 + 4i)^2 \\
&= 3^2 + 2(3)(4i) + (4i)^2 \\
&= 9 + 24i + 16i^2 \\
&= 9 + 24i - 16 \\
&= -7 + 24i
\end{aligned}
$$

Question 2f:
Given $w = 3 + 4i$. Let $\sqrt{w} = x + iy$.

$$(x + iy)^2 = 3 + 4i$$
$$x^2 - y^2 + 2xyi = 3 + 4i$$

Equating real and imaginary parts:

$$x^2 - y^2 = 3 \quad (1)$$
$$2xy = 4 \implies xy = 2 \quad (2)$$

From (2), $y = \frac{2}{x}$. Substitute into (1): $\text{x}^2 - \left(\frac{2}{x}\right)^2 = 3$

$x^2 - \frac{4}{x^2} = 3$
$(x^2)^2 - 3x^2 - 4 = 0$

Let $u = x^2$. Then $u^2 - 3u - 4 = 0 \implies (u-4)(u+1) = 0$. Since x is real, $x^2 = 4$, so $x = \pm 2$.
If $x = 2$, then $y = \frac{2}{2} = 1$. If $x = -2$, then $y = \frac{2}{-2} = -1$.
Thus, $\sqrt{w} = \pm(2 + i)$.

Question 2g:
Given $z = 6 - 7i$.

$$\bar{z} = 6 + 7i$$
$$\text{Re}(\bar{z}) = 6$$

Question 2h:
Given $w = 3 + 4i$.

$$\bar{w} = 3 - 4i$$
$$\text{Im}(\bar{w}) = -4$$

Question 2i:
Given $z = 6 - 7i$.

$$|z| = \sqrt{6^2 + (-7)^2}$$
$$= \sqrt{36 + 49}$$
$$= \sqrt{85}$$

Question 2j:
Given $w = 3 + 4i$.

$$
\begin{aligned}
|w| &= \sqrt{3^2 + 4^2} \\
&= \sqrt{9 + 16} \\
&= \sqrt{25} \\
&= 5
\end{aligned}
$$

Question 3a:
The equation is $z^2 + 5z + 1 = 0$. This is a quadratic equation in z.

$$
\begin{aligned}
\text{Degree of polynomial} &= 2 \\
\text{Number of roots} &= 2
\end{aligned}
$$

Since the discriminant $\Delta = 5^2 - 4(1)(1) = 21 > 0$, the roots are distinct and real.

Question 3b:
The equation is $2z^3 + 6z + 7 = 0$. This is a cubic equation in z.

$$
\begin{aligned}
\text{Degree of polynomial} &= 3 \\
\text{Number of roots} &= 3
\end{aligned}
$$

By the Fundamental Theorem of Algebra, a polynomial of degree n has exactly n roots in the complex plane (counting multiplicity). For a cubic with real coefficients, there is at least one real root and two other roots (which could be complex conjugates).

Question 3c:
The equation is $(3z^2 + 4z + 5)(z^2 + i) = 0$.

$$
\begin{aligned}
\text{Factor 1: } & 3z^2 + 4z + 5 = 0 \\
\text{Discriminant } & \Delta_1 = 4^2 - 4(3)(5) = 16 - 60 = -44 \\
\text{Roots of Factor 1: } & \text{2 distinct complex roots} \\
\text{Factor 2: } & z^2 + i = 0 \\
\text{Roots of Factor 2: } & z^2 = -i \implies \text{2 distinct complex roots} \\
\text{Total roots: } & 2 + 2 = 4
\end{aligned}
$$

There are 4 distinct roots.

Question 3d:
The equation is $z^5 + 5i = 0$.

$$
\begin{aligned}
\text{Degree of polynomial} &= 5 \\
\text{Number of roots} &= 5
\end{aligned}
$$

Since $z^5 = -5i$, there are 5 distinct 5th roots of $-5i$ in the complex plane.

Question 4:
A quadratic polynomial with real coefficients has roots that occur in conjugate pairs if they are complex.

$$\text{Given root: } z_1 = 1 + i$$
$$\text{Conjugate root: } z_2 = \bar{z}_1 = 1 - i$$

Therefore, the other root is $1 - i$.

Skill-building

Question 1a:
Solve $2z^2 - 2z + 5 = 0$.

$$\text{Using the quadratic formula: } z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$\begin{aligned} z &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(2)(5)}}{2(2)} \\ &= \frac{2 \pm \sqrt{4 - 40}}{4} \\ &= \frac{2 \pm \sqrt{-36}}{4} \\ &= \frac{2 \pm 6i}{4} \\ &= \frac{1}{2} \pm \frac{3}{2}i \end{aligned}$$

Question 1b:
Solve $5z^2 + 2z + 1 = 0$.

$$\begin{aligned} z &= \frac{-2 \pm \sqrt{2^2 - 4(5)(1)}}{2(5)} \\ &= \frac{-2 \pm \sqrt{4 - 20}}{10} \\ &= \frac{-2 \pm \sqrt{-16}}{10} \\ &= \frac{-2 \pm 4i}{10} \\ &= -\frac{1}{5} \pm \frac{2}{5}i \end{aligned}$$

Question 1c:
Solve $z^2 + z + 3 = 0$.

$$
\begin{aligned}
z &= \frac{-1 \pm \sqrt{1^2 - 4(1)(3)}}{2(1)} \\
&= \frac{-1 \pm \sqrt{1 - 12}}{2} \\
&= \frac{-1 \pm \sqrt{-11}}{2} \\
&= -\frac{1}{2} \pm \frac{\sqrt{11}}{2} i
\end{aligned}
$$

Question 2:
Given $z + \bar{z} = 12$ and $\bar{z} - z = 4i$.

Let $z = x + iy$, then $\bar{z} = x - iy$.

$$
\begin{aligned}
(x + iy) + (x - iy) &= 12 \\
2x &= 12 \\
\implies x &= 6
\end{aligned}
$$

Now use the second equation:

$$
\begin{aligned}
(x - iy) - (x + iy) &= 4i \\
-2iy &= 4i \\
\implies -2y &= 4 \\
\implies y &= -2
\end{aligned}
$$

Thus, $z = 6 - 2i$.

Question 3a:
Let $z = x + iy$, then $\bar{z} = x - iy$.

$$
\begin{aligned}
\frac{z + \bar{z}}{2} &= \frac{(x + iy) + (x - iy)}{2} \\
&= \frac{2x}{2} \\
&= x \\
&= \mathrm{Re}(z)
\end{aligned}
$$

Question 3b:
Let $z = x + iy$, then $\bar{z} = x - iy$.

$$
\begin{aligned}
\frac{z - \bar{z}}{2i} &= \frac{(x + iy) - (x - iy)}{2i} \\
&= \frac{2iy}{2i} \\
&= y \\
&= \mathrm{Im}(z)
\end{aligned}
$$

Question 3c:
Let $z = x + iy$, then $\bar{z} = x - iy$.

$$\begin{aligned} z\bar{z} &= (x+iy)(x-iy) \\ &= x^2 - (iy)^2 \\ &= x^2 - i^2y^2 \\ &= x^2 - (-1)y^2 \\ &= x^2 + y^2 \\ \text{Since } |z| = \sqrt{x^2+y^2}&\text{, then } |z|^2 = x^2 + y^2 \end{aligned}$$

Therefore, $|z|^2 = z\bar{z}$.

Question 4a:
Given $z^3 - 4z^2 + 9z - 10 = 0$ and one root $z_1 = 1 + 2i$.
Since the coefficients are real, the conjugate $z_2 = 1 - 2i$ must also be a root.

$$\begin{aligned} \text{Quadratic factor: } &(z - (1+2i))(z - (1-2i)) \\ &= ((z-1) - 2i)((z-1) + 2i) \\ &= (z-1)^2 - (2i)^2 \\ &= z^2 - 2z + 1 - (-4) \\ &= z^2 - 2z + 5 \end{aligned}$$

Now, divide the polynomial by $z^2 - 2z + 5$:

$$\begin{aligned} z^3 - 4z^2 + 9z - 10 &= (z^2 - 2z + 5)(z - 2) \\ \text{Check: } (z^2 - 2z + 5)(z-2) &= z^3 - 2z^2 - 2z^2 + 4z + 5z - 10 \\ &= z^3 - 4z^2 + 9z - 10 \end{aligned}$$

Thus, the roots are $z = 1 + 2i, 1 - 2i, 2$.

Question 4b:
Given $z^3 - 7z^2 + 17z - 15 = 0$ and one root $z_1 = 2 - i$.
Since the coefficients are real, the conjugate $z_2 = 2 + i$ must also be a root.

$$\begin{aligned} \text{Quadratic factor: } &(z - (2-i))(z - (2+i)) \\ &= ((z-2) + i)((z-2) - i) \\ &= (z-2)^2 - i^2 \\ &= z^2 - 4z + 4 - (-1) \\ &= z^2 - 4z + 5 \end{aligned}$$

Now, divide the polynomial by $z^2 - 4z + 5$:

$$\begin{aligned} z^3 - 7z^2 + 17z - 15 &= (z^2 - 4z + 5)(z - 3) \\ \text{Check: } (z^2 - 4z + 5)(z-3) &= z^3 - 3z^2 - 4z^2 + 12z + 5z - 15 \\ &= z^3 - 7z^2 + 17z - 15 \end{aligned}$$

Thus, the roots are $z = 2 - i, 2 + i, 3$.

Question 4c:
Given $z^4 - 6z^3 + 15z^2 - 18z + 10 = 0$ and one root $z_1 = 1 + i$.
Since the coefficients are real, the conjugate $z_2 = 1 - i$ must also be a root.

$$\begin{aligned}
\text{Quadratic factor: } & (z - (1+i))(z - (1-i)) \\
&= ((z-1) - i)((z-1) + i) \\
&= (z-1)^2 - i^2 \\
&= z^2 - 2z + 1 + 1 \\
&= z^2 - 2z + 2
\end{aligned}$$

Now, divide the polynomial by $z^2 - 2z + 2$:

$$z^4 - 6z^3 + 15z^2 - 18z + 10 = (z^2 - 2z + 2)(z^2 - 4z + 5)$$

Now solve $z^2 - 4z + 5 = 0$:

$$\begin{aligned}
z &= \frac{4 \pm \sqrt{16 - 20}}{2} \\
&= \frac{4 \pm \sqrt{-4}}{2} \\
&= \frac{4 \pm 2i}{2} \\
&= 2 \pm i
\end{aligned}$$

Thus, the roots are $z = 1 + i, 1 - i, 2 + i, 2 - i$.

Question 4d:
Given $z^4 - 4z^3 + 10z^2 - 12z + 5 = 0$ and one root $z_1 = 1 - 2i$.
Since the coefficients are real, the conjugate $z_2 = 1 + 2i$ must also be a root.

$$\begin{aligned}
\text{Quadratic factor: } & (z - (1-2i))(z - (1+2i)) \\
&= ((z-1) + 2i)((z-1) - 2i) \\
&= (z-1)^2 - (2i)^2 \\
&= z^2 - 2z + 1 + 4 \\
&= z^2 - 2z + 5
\end{aligned}$$

Now, divide the polynomial by $z^2 - 2z + 5$:

$$z^4 - 4z^3 + 10z^2 - 12z + 5 = (z^2 - 2z + 5)(z^2 - 2z + 1)$$

Now solve $z^2 - 2z + 1 = 0$:

$$\begin{aligned}
(z-1)^2 &= 0 \\
\implies z &= 1 \text{ (double root)}
\end{aligned}$$

Thus, the roots are $z = 1 - 2i, 1 + 2i, 1, 1$.

Question 5a:
Given $z^3 + 3z^2 + 7z + 5 = 0$ and one root $z_1 = -1 + 2i$.
Since the coefficients are real, the conjugate $z_2 = -1 - 2i$ must also be a root.

$$\begin{aligned}\text{Quadratic factor: } & (z - (-1 + 2i))(z - (-1 - 2i)) \\ &= (z + 1 - 2i)(z + 1 + 2i) \\ &= (z + 1)^2 - (2i)^2 \\ &= z^2 + 2z + 1 + 4 \\ &= z^2 + 2z + 5\end{aligned}$$

Now, divide the polynomial by $z^2 + 2z + 5$:

$$z^3 + 3z^2 + 7z + 5 = (z^2 + 2z + 5)(z + 1)$$

The factors with real coefficients are $(z + 1)(z^2 + 2z + 5)$.

Question 5b:
Given $z^3 - 2z^2 + 2z + 5 = 0$ and one root $z_1 = 1 - 2i$.
Since the coefficients are real, the conjugate $z_2 = 1 + 2i$ must also be a root.

$$\begin{aligned}\text{Quadratic factor: } & (z - (1 - 2i))(z - (1 + 2i)) \\ &= (z - 1 + 2i)(z - 1 - 2i) \\ &= (z - 1)^2 - (2i)^2 \\ &= z^2 - 2z + 1 + 4 \\ &= z^2 - 2z + 5\end{aligned}$$

Now, divide the polynomial by $z^2 - 2z + 5$:

$$z^3 - 2z^2 + 2z + 5 = (z^2 - 2z + 5)(z + 1)$$

The factors with real coefficients are $(z + 1)(z^2 - 2z + 5)$.

Question 5c:
Given $z^4 - 6z^3 + 23z^2 - 50z + 50 = 0$ and one root $z_1 = 1 - 3i$.
Since the coefficients are real, the conjugate $z_2 = 1 + 3i$ must also be a root.

$$\begin{aligned}\text{Quadratic factor: } & (z - (1 - 3i))(z - (1 + 3i)) \\ &= (z - 1 + 3i)(z - 1 - 3i) \\ &= (z - 1)^2 - (3i)^2 \\ &= z^2 - 2z + 1 + 9 \\ &= z^2 - 2z + 10\end{aligned}$$

Now, divide the polynomial by $z^2 - 2z + 10$:

$$z^4 - 6z^3 + 23z^2 - 50z + 50 = (z^2 - 2z + 10)(z^2 - 4z + 5)$$

Both quadratic factors have real coefficients. The factorisation is $(z^2 - 2z + 10)(z^2 - 4z + 5)$.

Question 5d:
Given $z^4 + 2z^3 - 2z^2 + 6z - 15 = 0$ and one root $z_1 = i\sqrt{3}$.
Since the coefficients are real, the conjugate $z_2 = -i\sqrt{3}$ must also be a root.

$$\begin{aligned}\text{Quadratic factor: } & (z - i\sqrt{3})(z + i\sqrt{3}) \\ &= z^2 - (i\sqrt{3})^2 \\ &= z^2 - (-3) \\ &= z^2 + 3\end{aligned}$$

Now, divide the polynomial by $z^2 + 3$:

$$z^4 + 2z^3 - 2z^2 + 6z - 15 = (z^2 + 3)(z^2 + 2z - 5)$$

The factors with real coefficients are $(z^2 + 3)(z^2 + 2z - 5)$.

Easier Exam Questions

Question 1i:
Given $z_1 = -1 + 3i$ and $z_2 = 2 - i$.

$$\begin{aligned}z_1 + z_2 &= (-1 + 3i) + (2 - i) \\ &= (-1 + 2) + (3 - 1)i \\ &= 1 + 2i\end{aligned}$$

Question 1ii:
Given $z_1 = -1 + 3i$ and $z_2 = 2 - i$.

$$\begin{aligned}z_1 z_2 &= (-1 + 3i)(2 - i) \\ &= -1(2) - 1(-i) + 3i(2) + 3i(-i) \\ &= -2 + i + 6i - 3i^2 \\ &= -2 + 7i + 3 \\ &= 1 + 7i\end{aligned}$$

Question 1iii:
Given $z_1 = -1 + 3i$ and $z_2 = 2 - i$.

$$
\begin{aligned}
\mathrm{Re}(z_1) &= -1 \\
\mathrm{Im}(z_2) &= -1 \\
\mathrm{Re}(z_1) - \mathrm{Im}(z_2) &= -1 - (-1) \\
&= -1 + 1 \\
&= 0
\end{aligned}
$$

Question 2i:
Given $z_1 = 2 - 3i$ is a root of $z^3 + mz + 52 = 0$ with $m \in \mathbb{R}$.

Since the coefficients are real, the conjugate $z_2 = 2 + 3i$ must also be a root.

$$
\begin{aligned}
\text{Quadratic factor: } & (z - (2 - 3i))(z - (2 + 3i)) \\
&= (z - 2 + 3i)(z - 2 - 3i) \\
&= (z - 2)^2 - (3i)^2 \\
&= z^2 - 4z + 4 + 9 \\
&= z^2 - 4z + 13
\end{aligned}
$$

Let the third root be $z_3 = r$. Since the polynomial is $z^3 + 0z^2 + mz + 52 = 0$, the sum of roots is:

$$
\begin{aligned}
z_1 + z_2 + z_3 &= -\frac{\text{coefficient of } z^2}{\text{coefficient of } z^3} \\
(2 - 3i) + (2 + 3i) + r &= 0 \\
4 + r &= 0 \\
\implies r &= -4
\end{aligned}
$$

Thus, the other roots are $2 + 3i$ and -4.

Question 2ii:
From part (i), the roots are $z_1 = 2 - 3i, z_2 = 2 + 3i, z_3 = -4$.

$$
\begin{aligned}
\text{Product of roots: } z_1 z_2 z_3 &= -\frac{\text{constant term}}{\text{coefficient of } z^3} \\
(2 - 3i)(2 + 3i)(-4) &= -52 \\
(4 + 9)(-4) &= -52 \\
13(-4) &= -52 \\
-52 &= -52 \quad \text{(Correct)}
\end{aligned}
$$

To find m, we use the sum of roots taken two at a time:

$$
\begin{aligned}
\sum z_i z_j &= \frac{\text{coefficient of } z}{\text{coefficient of } z^3} \\
(2 - 3i)(2 + 3i) + (2 - 3i)(-4) + (2 + 3i)(-4) &= m \\
13 + (-8 + 12i) + (-8 - 12i) &= m \\
13 - 16 &= m \\
\implies m &= -3
\end{aligned}
$$

Question 3i:
Given $z = 3 - i$ and $w = 2 + 5i$.

$$\begin{aligned} z + w &= (3 - i) + (2 + 5i) \\ &= (3 + 2) + (-1 + 5)i \\ &= 5 + 4i \end{aligned}$$

$$\begin{aligned} |z + w| &= \sqrt{5^2 + 4^2} \\ &= \sqrt{25 + 16} \\ &= \sqrt{41} \end{aligned}$$

Question 3ii:
Given $z = 3 - i$ and $w = 2 + 5i$.

$$\begin{aligned} \frac{w}{z} &= \frac{2 + 5i}{3 - i} \times \frac{3 + i}{3 + i} \\ &= \frac{(2 + 5i)(3 + i)}{3^2 + (-1)^2} \\ &= \frac{6 + 2i + 15i + 5i^2}{9 + 1} \\ &= \frac{6 + 17i - 5}{10} \\ &= \frac{1 + 17i}{10} \\ &= 0.1 + 1.7i \end{aligned}$$

Question 4:
Find the square roots of $3 + 2i\sqrt{10}$. Let $\sqrt{3 + 2i\sqrt{10}} = x + iy$.

$$\begin{aligned} (x + iy)^2 &= 3 + 2i\sqrt{10} \\ x^2 - y^2 + 2xyi &= 3 + 2i\sqrt{10} \end{aligned}$$

Equating real and imaginary parts:

$$\begin{aligned} x^2 - y^2 &= 3 \quad (1) \\ 2xy &= 2\sqrt{10} \implies xy = \sqrt{10} \quad (2) \end{aligned}$$

From (2), $y = \frac{\sqrt{10}}{x}$. Substitute into (1):

$$\begin{aligned} x^2 - \left(\frac{\sqrt{10}}{x}\right)^2 &= 3 \\ x^2 - \frac{10}{x^2} &= 3 \\ (x^2)^2 - 3x^2 - 10 &= 0 \end{aligned}$$

Let $u = x^2$. Then $u^2 - 3u - 10 = 0 \implies (u-5)(u+2) = 0$. Since x is real, $x^2 = 5$, so $x = \pm\sqrt{5}$.

If $x = \sqrt{5}$, then $y = \frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2}$. If $x = -\sqrt{5}$, then $y = \frac{\sqrt{10}}{-\sqrt{5}} = -\sqrt{2}$.

Thus, the square roots are $\pm(extsqrt5 + i\text{sqrt}2)$.

Question 5i:
Given $z = 2 + 3i$ and $w = 1 - 5i$.

$$\begin{aligned}
\bar{w} &= 1 + 5i \\
z + \bar{w} &= (2 + 3i) + (1 + 5i) \\
&= (2 + 1) + (3 + 5)i \\
&= 3 + 8i
\end{aligned}$$

Question 5ii:
Given $z = 2 + 3i$.

$$\begin{aligned}
z^2 &= (2 + 3i)^2 \\
&= 2^2 + 2(2)(3i) + (3i)^2 \\
&= 4 + 12i + 9i^2 \\
&= 4 + 12i - 9 \\
&= -5 + 12i
\end{aligned}$$

Question 6:
The equation is $(z^2 - 2zi - 1)(z^2 + 2i)(z^2 + 2zi + 2) = 0$.

Factor 1: $z^2 - 2zi - 1 = 0$

$$\begin{aligned}
(z - i)^2 &= 0 \\
\implies z = i \text{ (double root)}
\end{aligned}$$

Factor 2: $z^2 + 2i = 0$

$$\begin{aligned}
z^2 &= -2i \\
\text{Let } z = x + iy \implies x^2 - y^2 + 2xyi &= -2i \\
\implies x^2 - y^2 = 0 \text{ and } 2xy &= -2 \\
\implies x = \pm y \text{ and } xy &= -1 \\
\text{If } x = y, x^2 = -1 \text{ (impossible for real } x) \\
\text{If } x = -y, -x^2 = -1 \implies x &= \pm 1 \\
\text{If } x = 1, y = -1 \implies z &= 1 - i \\
\text{If } x = -1, y = 1 \implies z &= -1 + i
\end{aligned}$$

Factor 3: $z^2 + 2zi + 2 = 0$

$$\begin{aligned}
\text{Using quadratic formula: } z &= \frac{-2i \pm \sqrt{(2i)^2 - 4(1)(2)}}{2} \\
&= \frac{-2i \pm \sqrt{-4-8}}{2} \\
&= \frac{-2i \pm \sqrt{-12}}{2} \\
&= \frac{-2i \pm 2i\sqrt{3}}{2} \\
&= -i \pm i\sqrt{3} = i(-1 \pm \sqrt{3})
\end{aligned}$$

Distinct roots are: $i, 1-i, -1+i, i(-1+\sqrt{3}), i(-1-\sqrt{3})$.
There are 5 distinct roots. Correct option is C.

Question 7:
A cubic polynomial $P(z)$ with real coefficients has roots $z_1 = 3$ and $z_2 = 2+i$.
Since the coefficients are real, the conjugate $z_3 = 2-i$ must also be a root.

$$\begin{aligned}
\text{Quadratic factor: } &(z-(2+i))(z-(2-i)) \\
&= (z-2-i)(z-2+i) \\
&= (z-2)^2 - i^2 \\
&= z^2 - 4z + 4 + 1 \\
&= z^2 - 4z + 5
\end{aligned}$$

Now, multiply by the linear factor $(z-3)$:

$$\begin{aligned}
P(z) &= (z-3)(z^2-4z+5) \\
&= z(z^2-4z+5) - 3(z^2-4z+5) \\
&= z^3 - 4z^2 + 5z - 3z^2 + 12z - 15 \\
&= z^3 - 7z^2 + 17z - 15
\end{aligned}$$

Comparing with the options, the correct option is C.

Harder Exam Questions

Question 1i:
To show $\sqrt{2}-i$ is a root of $z^3 - (\sqrt{2}-i)z^2 + 8z - 8\sqrt{2} + 8i = 0$, we substitute $z = \sqrt{2}-i$ into the polynomial.

$$\begin{aligned}
\text{Let } P(z) &= z^3 - (\sqrt{2}-i)z^2 + 8z - 8\sqrt{2} + 8i \\
P(\sqrt{2}-i) &= (\sqrt{2}-i)^3 - (\sqrt{2}-i)(\sqrt{2}-i)^2 + 8(\sqrt{2}-i) - 8\sqrt{2} + 8i \\
&= (\sqrt{2}-i)^3 - (\sqrt{2}-i)^3 + 8\sqrt{2} - 8i - 8\sqrt{2} + 8i \\
&= 0
\end{aligned}$$

Since $P(\sqrt{2}-i) = 0$, $\sqrt{2}-i$ is a root.

Question 1ii:
Given the root $z_1 = \sqrt{2} - i$. The polynomial is $z^3 - (\sqrt{2} - i)z^2 + 8z - 8\sqrt{2} + 8i = 0$.

We can factor the polynomial by grouping:

$$\begin{aligned} P(z) &= z^2(z - (\sqrt{2} - i)) + 8(z - (\sqrt{2} - i)) \\ &= (z^2 + 8)(z - (\sqrt{2} - i)) \end{aligned}$$

To find the other two roots, solve $z^2 + 8 = 0$:

$$\begin{aligned} z^2 &= -8 \\ z &= \pm\sqrt{-8} \\ &= \pm 2i\sqrt{2} \end{aligned}$$

Thus, the other two solutions are $z = 2i\sqrt{2}$ and $z = -2i\sqrt{2}$.

Question 2i:
Find the square roots of $-3 - 4i$. Let $\sqrt{-3 - 4i} = x + iy$.

$$\begin{aligned} (x + iy)^2 &= -3 - 4i \\ x^2 - y^2 + 2xyi &= -3 - 4i \end{aligned}$$

Equating real and imaginary parts:

$$\begin{aligned} x^2 - y^2 &= -3 \quad (1) \\ 2xy &= -4 \implies xy = -2 \quad (2) \end{aligned}$$

From (2), $y = -\frac{2}{x}$. Substitute into (1):

$$\begin{aligned} x^2 - \left(-\frac{2}{x}\right)^2 &= -3 \\ x^2 - \frac{4}{x^2} &= -3 \\ (x^2)^2 + 3x^2 - 4 = 0 \end{aligned}$$

Let $u = x^2$. Then $u^2 + 3u - 4 = 0 \implies (u + 4)(u - 1) = 0$. Since x is real, $x^2 = 1$, so $x = \pm 1$.
If $x = 1$, then $y = -\frac{2}{1} = -2$. If $x = -1$, then $y = -\frac{2}{-1} = 2$.
Thus, the square roots are $\pm(1 - 2i)$.

Question 2ii:
Solve $z^2 - 3z + (3 + i) = 0$.

$$\begin{aligned} \text{Using the quadratic formula: } z &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ z &= \frac{3 \pm \sqrt{(-3)^2 - 4(1)(3 + i)}}{2} \\ &= \frac{3 \pm \sqrt{9 - 12 - 4i}}{2} \\ &= \frac{3 \pm \sqrt{-3 - 4i}}{2} \end{aligned}$$

From part (i), $\sqrt{-3-4i} = \pm(1-2i)$.

$$\begin{aligned}
z &= \frac{3 \pm (1-2i)}{2} \\
\text{Case 1: } z &= \frac{3+1-2i}{2} = \frac{4-2i}{2} = 2-i \\
\text{Case 2: } z &= \frac{3-1+2i}{2} = \frac{2+2i}{2} = 1+i
\end{aligned}$$

Thus, the solutions are $z = 2-i$ and $z = 1+i$.

Question 3i:
Find the square root(s) of $7-24i$. Let $\sqrt{7-24i} = x+iy$.

$$\begin{aligned}
(x+iy)^2 &= 7-24i \\
x^2 - y^2 + 2xyi &= 7-24i
\end{aligned}$$

Equating real and imaginary parts:

$$\begin{aligned}
x^2 - y^2 &= 7 \quad (1) \\
2xy &= -24 \implies xy = -12 \quad (2)
\end{aligned}$$

From (2), $y = -\frac{12}{x}$. Substitute into (1):

$$\begin{aligned}
x^2 - \left(-\frac{12}{x}\right)^2 &= 7 \\
x^2 - \frac{144}{x^2} &= 7 \\
(x^2)^2 - 7x^2 - 144 = 0
\end{aligned}$$

Let $u = x^2$. Then $u^2 - 7u - 144 = 0 \implies (u-16)(u+9) = 0$. Since x is real, $x^2 = 16$, so $x = \pm 4$.

If $x = 4$, then $y = -\frac{12}{4} = -3$. If $x = -4$, then $y = -\frac{12}{-4} = 3$.

Thus, the square roots are $\pm(4-3i)$.

Question 3ii:
Solve $2z^2 + 6z + (1+12i) = 0$.

$$\begin{aligned}
\text{Using the quadratic formula: } z &= \frac{-6 \pm \sqrt{6^2 - 4(2)(1+12i)}}{2(2)} \\
&= \frac{-6 \pm \sqrt{36-8-96i}}{2(2)} \\
&= \frac{-6 \pm \sqrt{28-96i}}{4} \\
&= \frac{-6 \pm \sqrt{4(7-24i)}}{4} \\
&= \frac{-6 \pm 2\sqrt{7-24i}}{4}
\end{aligned}$$

From part (i), $\sqrt{7-24i} = \pm(4-3i)$.

$$\begin{aligned} z &= \frac{-6 \pm 2(4-3i)}{4} \\ \text{Case 1: } z &= \frac{-6+8-6i}{4} = \frac{2-6i}{4} = 0.5 - 1.5i \\ \text{Case 2: } z &= \frac{-6-8+6i}{4} = \frac{-14+6i}{4} = -3.5 + 1.5i \end{aligned}$$

Question 4i:
Given $P(x) = x^4 + ax^3 - 40x^2 + 41x + b$ with $a, b \in \mathbb{R}$.

One zero is $x = \frac{1-i\sqrt{3}}{2}$. Since the coefficients are real, the conjugate $x = \frac{1+i\sqrt{3}}{2}$ must also be a zero.

$$\begin{aligned} \text{Quadratic factor: } &(x - \frac{1-i\sqrt{3}}{2})(x - \frac{1+i\sqrt{3}}{2}) \\ &= (x - \frac{1}{2} + \frac{i\sqrt{3}}{2})(x - \frac{1}{2} - \frac{i\sqrt{3}}{2}) \\ &= (x - \frac{1}{2})^2 - (\frac{i\sqrt{3}}{2})^2 \\ &= x^2 - x + \frac{1}{4} - (-\frac{3}{4}) \\ &= x^2 - x + 1 \end{aligned}$$

Since $x^2 - x + 1$ is formed by two zeros of $P(x)$, it must be a factor of $P(x)$.

Question 4ii:
We know $x = 7$ is a zero, so $(x-7)$ is a factor. We also know $(x^2 - x + 1)$ is a factor.

$$\begin{aligned} \text{Let } P(x) &= (x-7)(x^2-x+1)(x-r) \\ \text{Expanding the first two factors: } (x-7)(x^2-x+1) &= x^3 - x^2 + x - 7x^2 + 7x - 7 \\ &= x^3 - 8x^2 + 8x - 7 \end{aligned}$$

Now, $P(x) = (x^3 - 8x^2 + 8x - 7)(x - r)$.
Comparing the coefficient of x^4 (which is 1) and the constant term b:

$$\begin{aligned} \text{Constant term: } &-7(-r) = 7r = b \\ \text{Coefficient of } x^3: &-r - 8 = a \end{aligned}$$

Equating the coefficient of x^2 to -40:

$$\begin{aligned} 8r + 8 &= -40 \\ 8r &= -48 \\ \implies r &= -6 \end{aligned}$$

Now find a and b:

$$\begin{aligned} a &= -r - 8 = -(-6) - 8 = 6 - 8 = -2 \\ b &= 7r = 7(-6) = -42 \end{aligned}$$

Question 5:
Given $P(z) = z^4 - 2z^3 + az^2 + bz + 50$ with $a, b \in \mathbb{R}$ and $P(2-i) = 0$.
Since the coefficients are real, the conjugate $z = 2 + i$ must also be a root.

$$\begin{aligned} \text{Quadratic factor: } & (z - (2-i))(z - (2+i)) \\ &= (z - 2 + i)(z - 2 - i) \\ &= (z-2)^2 - i^2 \\ &= z^2 - 4z + 4 + 1 \\ &= z^2 - 4z + 5 \end{aligned}$$

Now, divide $P(z)$ by $z^2 - 4z + 5$:

$$\begin{aligned} \text{Let } P(z) &= (z^2 - 4z + 5)(z^2 + cz + d) \\ \text{Expanding: } (z^2 - 4z + 5)(z^2 + cz + d) &= z^4 + cz^3 + dz^2 - 4z^3 - 4cz^2 - 4dz + 5z^2 + 5cz + 5d \\ &= z^4 + (c-4)z^3 + (d - 4c + 5)z^2 + (5c - 4d)z + 5d \end{aligned}$$

Comparing coefficients with $P(z) = z^4 - 2z^3 + az^2 + bz + 50$:

$$\begin{aligned} \text{Coefficient of } z^3: c - 4 = -2 &\implies c = 2 \\ \text{Constant term: } 5d = 50 &\implies d = 10 \end{aligned}$$

Thus, the other quadratic factor is $z^2 + 2z + 10$.

$$\text{Factorisation: } P(z) = (z^2 - 4z + 5)(z^2 + 2z + 10)$$

Question 6:
Given $P(z) = z^4 - 6z^3 + 18z^2 - 14z - 39$ and one root $z_1 = 2 - 3i$.
Since the coefficients are real, the conjugate $z_2 = 2 + 3i$ must also be a root.

$$\begin{aligned} \text{Quadratic factor: } & (z - (2-3i))(z - (2+3i)) \\ &= (z - 2 + 3i)(z - 2 - 3i) \\ &= (z-2)^2 - (3i)^2 \\ &= z^2 - 4z + 4 + 9 \\ &= z^2 - 4z + 13 \end{aligned}$$

Now, divide $P(z)$ by $z^2 - 4z + 13$:

$$\text{Using polynomial division: } z^4 - 6z^3 + 18z^2 - 14z - 39 = (z^2 - 4z + 13)(z^2 - 2z - 3)$$

Now, factorise the remaining quadratic $z^2 - 2z - 3$:

$$z^2 - 2z - 3 = (z - 3)(z + 1)$$

Thus, the full factorisation is:

$$P(z) = (z^2 - 4z + 13)(z - 3)(z + 1)$$